

* Encapsulation :

- wrapping data (variables) and methods (functions).
- private, public, protected → (access modifiers).

* Abstraction :

- Essential features & hiding complex implementation details.
- abstraction is achieved using abstract keyword & Interfaces.

* Inheritance → to acquire properties & behaviors (fields & methods) from one to another class (parent to child class).

* Polymorphism : Many forms.

compiles poly. : ~~overloading~~ ^{overriding} method
runtime poly. : ~~overloading~~ ^{overriding} method

* Function : → block of code for specific actions.

user-defined ↓
predefined

→ specifically for reusing.

→ return statement type → value can be stored & used later.

* Multi-level Inheritance :

Base parent class



Derived class



More levels